Applied AI and Automation for Executives

14 sessions of 2 hours | 28 hours | dates and format agreed with you

Session 0 is an introduction with no activities: how AI and machine learning work, the fields and jobs behind them, and a handbook to read before Session 1. Sessions 1 to 11 open the box: how a machine sees, learns, is measured, reads words and predicts the next word, then foundation models, open weights, prompting, giving a model your own knowledge, failures and agents. I demonstrate each idea and you may join in. There is no coding. Sessions 12 to 14 are practical, and you work on your own organisation.

You understand how AI works without needing any maths or computer science background: every idea is taught as a picture, an everyday comparison, a calculation with at most three numbers, the real term, and a demonstration. By the end you have mapped one of your own workflows, written an AI-use policy and a vendor question list, assessed your organisation's data readiness, and built a data roadmap and a 90-day plan. Every session ends with a one-page take-away you keep. There are no tests and no marks. No coding experience is needed.

The fourteen sessions

No.	Session	You keep
1	Seeing: pixels, features and neurons	A one-page sheet: how a picture becomes numbers, a neuron worked by hand on a house price and an umbrella, and the digit results
2	Learning from examples: types of machine learning, decision trees, nearest neighbours	A one-page sheet: the four types of learning, a tree read as rules, k-nearest neighbours, and the honest scores from today's run
3	Measuring a model: accuracy, precision, recall, F1, ROC and AUC, thresholds, baselines, overfitting	The vendor question sheet: ten questions about the numbers behind any AI claim
4	Words as numbers: bag of words, TF-IDF, embeddings and distance	The cheat sheet of the four measures: Euclidean distance, Manhattan distance, dot product and cosine similarity, each with its formula, its picture and when to use it
5	Language models: tokens, attention and next-word probability	A one-page sheet: next-token prediction, how the model is trained, the softmax formula, what temperature, top-k and top-p do, self-attention with query, key and value, and five questions to ask about any language-model answer
6	Foundation models: pretraining, post-training, model families, reasoning and multimodal models, benchmarks	A one-page model comparison grid for your own work, with the questions to ask of a model card and a benchmark claim
7	Open weights, file formats, size and running cost	A one-page rent, host or own decision sheet, with the memory sum and the licence questions to ask
8	Prompting and reasoning models	A reusable prompt template and a prompt log, so that every prompt you keep has been tested on an evaluation set
9	Giving a model your knowledge: retrieval-augmented generation, long context and fine-tuning	A one-page decision tree: prompting first, then retrieval-augmented generation (RAG), then fine-tuning, and the test that tells you when to move on
10	When models fail, and how to check them	A five-step check routine and an evaluation-set template (questions with ground truth), each on one page
11	Agentic AI: tools, loops, MCP, autonomy and guardrails	Your Tool Boundary Sheet, the permissions table for an agent: for read, write, send and spend, what it may do, what a person approves (human-in-the-loop) and what it must never do, with the owner, the audit log and the off switch
12	Workflow mapping (practical)	Your mapped workflow, the choice for each step, your assistant sketch, and the baseline and hours-saved estimate in the Business Case Workbook

No.	Session	You keep
13	Governance and buying (practical)	Your Risk Register, your AI Use Policy draft, your Vendor Question List with five questions, and your page of questions for the lawyer
14	Data readiness assessment, roadmap and 90-day plan (practical)	Your inventory and scorecard, your roadmap with its risks and decisions, the Follow-up Kit (Check-in Form, Results Tracker, Sponsor Briefing) and your answer to the Session 0 question

What you will be able to do

- Explain in plain words how a machine sees, learns, reads words and predicts the next word, and what a foundation model, an open-weights model and an agent are.
- Judge a model with the right measures, and ask a vendor the questions that matter.
- Brief an AI tool well, give it your own knowledge, check what it returns, and know when not to trust it.
- Map a workflow and decide which steps need a rule, an AI step or an agent, and estimate the hours saved.
- Set the rules: an AI-use policy, a risk register, and the questions to put to any vendor.
- Assess your organisation's data readiness and lead a data roadmap and a 90-day plan.

How we work

Each session pairs short explanations with demonstrations that the instructor runs while you watch, and you may join in. Sessions 1 to 11 include a short paper exercise with no computer. Sessions 12 to 14 are practical and most of the time is yours. Examples come from a fictional company, Nile Valley Foods Ltd, and from finance, operations, customer service, HR, sales, compliance and legal settings, so you never need to share company data in class. Other tools on the market are discussed as options only.

Before the first session

- Nothing to prepare. Read the Course Handbook if you can.
- Bring a laptop and one AI assistant account you already use, if you want to join in the demonstrations.
- For Sessions 12 to 14, think of one manual workflow in your work, or the personal email you would like summarised.
- Use only sample or invented data for practice. Never paste personal or confidential data into a tool you have not approved.

After the course

You receive a follow-up kit. At 30, 60 and 90 days the instructor checks what is running, how much time it saves and what changed in your policy, and you return a one-page check-in form and a results tracker.

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